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- **Problem Statement:**

Hospital Management System[Array of Object/ Structure, Link List].

Project Title

HOSPITAL MANAGEMENT SYSTEM

- **Program:**

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
class Node {
```

```
public:
```

```
    int id;
```

```
    string name;
```

```
    int age;
```

```
    string disease;
```

```
    Node* next;
```

```
    Node(int id, string name, int age, string disease) {
```

```
        this->id = id;
```

```
        this->name = name;
```

```
        this->age = age;
```

```
        this->disease = disease;
```

```
        this->next = nullptr;
```

```
    }
```

```
};
```

```
class LinkedList {
```

```
private:
```

```
    Node* head;
```

```
public:
```

```
    LinkedList() {
```

```
        head = nullptr;
```

```
    }
```

```
    void addPatient(int id, string name, int age, string disease) {
```

```
        Node* newNode = new Node(id, name, age, disease);
```

```
        if (!head) head = newNode;
```

```
        else {
```

```
            Node* temp = head;
```

```
            while (temp->next) temp = temp->next;
```

```
            temp->next = newNode;
```

```
        }
```

```
        cout << "Patient added successfully!\n";
```

```
    }
```

```
    void displayPatients() {
```

```
        if (!head) {
```

```

        cout << "No patient records available.\n";

        return;
    }

    Node* temp = head;

    cout << "\n Patient List:\n";

    while (temp) {

        cout << "-----\n";

        cout << "ID    : " << temp->id << "\n";

        cout << "Name   : " << temp->name << "\n";

        cout << "Age    : " << temp->age << "\n";

        cout << "Disease : " << temp->disease << "\n";

        temp = temp->next;
    }

    cout << "-----\n";
}

```

```

void deletePatient(int id) {

    if (!head) {

        cout << "No records to delete.\n";

        return;
    }
}

```

```

Node* temp = head;

Node* prev = nullptr;

```

```
if (head->id == id) {  
    head = head->next;  
    delete temp;  
    cout << "Patient deleted successfully.\n";  
    return;  
}
```

```
while (temp && temp->id != id) {  
    prev = temp;  
    temp = temp->next;  
}
```

```
if (!temp) {  
    cout << "Patient with ID " << id << " not found.\n";  
    return;  
}
```

```
prev->next = temp->next;  
delete temp;  
cout << "Patient deleted successfully.\n";  
}  
};
```

```
class DoctorNode {  
public:  
    int id;  
    string name;  
    string specialization;  
    bool available;  
    DoctorNode* next;  
  
    DoctorNode(int id, string name, string specialization, bool available) {  
        this->id = id;  
        this->name = name;  
        this->specialization = specialization;  
        this->available = available;  
        this->next = nullptr;  
    }  
};
```

```
class DoctorList {  
private:  
    DoctorNode* head;  
  
public:  
    DoctorList() {  
        head = nullptr;  
    }  
};
```

```
}
```

```
void addDoctor(int id, string name, string specialization, bool available) {  
    DoctorNode* newNode = new DoctorNode(id, name, specialization, available);  
    if (!head) head = newNode;  
    else {  
        DoctorNode* temp = head;  
        while (temp->next) temp = temp->next;  
        temp->next = newNode;  
    }  
    cout << "Doctor added successfully!\n";  
}
```

```
void displayDoctors() {  
    if (!head) {  
        cout << "No doctors available.\n";  
        return;  
    }  
  
    DoctorNode* temp = head;  
    cout << "\nDoctor List:\n";  
    while (temp) {  
        cout << "-----\n";  
        cout << "ID          : " << temp->id << "\n";
```

```
        cout << "Name          : " << temp->name << "\n";

        cout << "Specialization : " << temp->specialization << "\n";

        cout << "Available     : " << (temp->available ? "Yes" : "No") << "\n";

        temp = temp->next;

    }

    cout << "-----\n";

}

};
```

```
class BedManager {

private:

    int totalBeds;

    int occupiedBeds;

public:

    BedManager(int total) {

        totalBeds = total;

        occupiedBeds = 0;

    }

    void admitPatient() {

        if (occupiedBeds < totalBeds) {

            occupiedBeds++;

            cout << "Patient admitted. Bed assigned.\n";
```



```
    } else {  
        cout << "No beds available!\n";  
    }  
}
```

```
void dischargePatient() {  
    if (occupiedBeds > 0) {  
        occupiedBeds--;  
        cout << "Patient discharged. Bed freed.\n";  
    } else {  
        cout << "No patients to discharge.\n";  
    }  
}
```

```
void displayBedStatus() {  
    cout << "\nBed Availability:\n";  
    cout << "Total Beds    : " << totalBeds << "\n";  
    cout << "Occupied Beds : " << occupiedBeds << "\n";  
    cout << "Available Beds : " << (totalBeds - occupiedBeds) << "\n";  
}  
};
```

```
int main() {  
    LinkedList patients;
```

```
DoctorList doctors;

BedManager beds(10);


int choice;


do {

    cout << "\nHospital Management System\n";

    cout << "1. Add Patient\n";

    cout << "2. Display Patients\n";

    cout << "3. Delete Patient\n";

    cout << "4. Add Doctor\n";

    cout << "5. Display Doctors\n";

    cout << "6. Admit Patient (Bed)\n";

    cout << "7. Discharge Patient (Bed)\n";

    cout << "8. Check Bed Availability\n";

    cout << "9. Exit\n" << endl;

    cout << "Enter your choice: ";

    cin >> choice;


    int id, age;

    string name, disease, specialization;

    bool available;


    switch (choice) {
```

case 1:

```
cout << "Enter Patient ID: ";  
  
cin >> id;  
  
cout << "Enter Name: ";  
  
cin >> name;  
  
cout << "Enter Age: ";  
  
cin >> age;  
  
cout << "Enter Disease: ";  
  
cin >> disease;  
  
patients.addPatient(id, name, age, disease);  
  
break;
```

case 2:

```
patients.displayPatients();  
  
break;
```

case 3:

```
cout << "Enter Patient ID to delete: ";  
  
cin >> id;  
  
patients.deletePatient(id);  
  
break;
```

case 4:

```
cout << "Enter Doctor ID: ";
```

```
cin >> id;

cout << "Enter Name: ";

cin >> name;

cout << "Enter Specialization: ";

cin >> specialization;

cout << "Is Doctor Available (1 for Yes / 0 for No): ";

cin >> available;

doctors.addDoctor(id, name, specialization, available);

break;
```

case 5:

```
doctors.displayDoctors();

break;
```

case 6:

```
beds.admitPatient();

break;
```

case 7:

```
beds.dischargePatient();

break;
```

case 8:

```
beds.displayBedStatus();
```

```
break;
```

```
case 9:
```

```
cout << "Exiting system. Stay healthy!\n";
```

```
break;
```

```
default:
```

```
cout << "Invalid choice. Try again.\n";
```

```
}
```

```
} while (choice != 9);
```

```
return 0;
```

```
}
```

• Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
powershell + - [ ] [ ] ... ^ x

PS C:\Users\editi\Documents\Gangotri\B.TECH\4th semester\VSEC-DS\Programs\Practicals> g++ Hospital.cpp
PS C:\Users\editi\Documents\Gangotri\B.TECH\4th semester\VSEC-DS\Programs\Practicals> ./a.exe

Hospital Management System
1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 1
Enter Patient ID: 1
Enter Name: Rohan
Enter Age: 23
Enter Disease: LungCancer
Patient added successfully!

Hospital Management System
1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
powershell + - [ ] [ ] ... ^ x

Enter your choice: 2

Patient List:
-----
ID       : 1
Name     : Rohan
Age      : 23
Disease  : LungCancer
-----

Hospital Management System
1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 1
Enter Patient ID: 2
Enter Name: Dhanraj
Enter Age: 21
Enter Disease: Diabetes
Patient added successfully!
```

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 1

Enter Patient ID: 3

Enter Name: Sakshi

Enter Age: 24

Enter Disease: Ectopicpregnency

Patient added successfully!

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 1

Enter Patient ID: 5

Enter Name: Neha

Enter Age: 22

Enter Disease: Kidneyfailure

Patient added successfully!

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 1

Enter Patient ID: 4

Enter Name: Shivam

Enter Age: 25

Enter Disease: Asthama

Patient added successfully!

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 2

Enter your choice: 2

Patient List:

```
-----
ID      : 1
Name    : Rohan
Age     : 23
Disease : LungCancer
-----
ID      : 2
Name    : Dhanraj
Age     : 21
Disease : Diabetes
-----
ID      : 3
Name    : Sakshi
Age     : 24
Disease : Ectopicpregnency
-----
ID      : 5
Name    : Neha
Age     : 22
Disease : Kidneyfailure
-----
ID      : 4
Name    : Shivam
Age     : 25
Disease : Asthama
-----
```

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

```
Enter your choice: 4
Enter Doctor ID: 1
Enter Name: Dr.Abhishekh
Enter Specialization: diabetoogist
Is Doctor Available (1 for Yes / 0 for No): 1
Doctor added successfully!
```

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

```
Enter your choice: 4
Enter Doctor ID: 2
Enter Name: Dr.hitesh
Enter Specialization: surgon
Is Doctor Available (1 for Yes / 0 for No): 1
Doctor added successfully!
```

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

```
Enter your choice: 4
Enter Doctor ID: 3
Enter Name: Dr.prathmesh
Enter Specialization: cardiologist
Is Doctor Available (1 for Yes / 0 for No): 1
Doctor added successfully!
```

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

```
Enter your choice: 4
Enter Doctor ID: 4
Enter Name: Dr.gangotri
Enter Specialization: gynecologist
Is Doctor Available (1 for Yes / 0 for No): 1
Doctor added successfully!
```

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

```
Enter your choice: 4
Enter Doctor ID: 5
Enter Name: Dr.kapil
Enter Specialization: neurologist
Is Doctor Available (1 for Yes / 0 for No): 1
Doctor added successfully!
```

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

```
Enter your choice: 5
```

Doctor List:

```
-----
ID           : 1
Name         : Dr.Abhishekh
Specialization : diabetoogist
Available    : Yes
-----
ID           : 2
Name         : Dr.hitesh
Specialization : surgon
Available    : Yes
-----
ID           : 3
Name         : Dr.prathmesh
Specialization : cardiologist
Available    : Yes
```

```
-----  
ID           : 4  
Name         : Dr.gangotri  
Specialization : gynecologist  
Available    : Yes  
-----
```

```
ID           : 5  
Name         : Dr.kapil  
Specialization : neurologist  
Available    : Yes  
-----
```

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 6
Patient admitted. Bed assigned.

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 6
Patient admitted. Bed assigned.

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 6

Patient admitted. Bed assigned.

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 7

Patient discharged. Bed freed.

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

```
Enter your choice: 3
Enter Patient ID to delete: 2
Patient deleted successfully.
```

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

```
Enter your choice: 2
```

Patient List:

```
-----
ID      : 1
Name    : Rohan
Age     : 23
Disease : LungCancer
-----
ID      : 3
Name    : Sakshi
Age     : 24
Disease : Ectopicpregnancy
-----
ID      : 5
Name    : Neha
Age     : 22
Disease : Kidneyfailure
-----
ID      : 4
Name    : Shivam
Age     : 25
Disease : Asthama
-----
```

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 8

Bed Availability:

Total Beds : 10
Occupied Beds : 2
Available Beds : 8

Hospital Management System

1. Add Patient
2. Display Patients
3. Delete Patient
4. Add Doctor
5. Display Doctors
6. Admit Patient (Bed)
7. Discharge Patient (Bed)
8. Check Bed Availability
9. Exit

Enter your choice: 9

Exiting system. Stay healthy!